Cubic Time Dilation Model for Enhanced GNSS

Second edition, revised: what the cube-root exponent actually explains

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Abstract

The first edition of this proposal (2024) replaced the standard relativistic time-dilation exponent, $-\frac{1}{2}$, with a cube-root exponent, $-\frac{1}{3}$, and reported 1,000 simulations validating it as more accurate for GNSS. Three independent lines of evidence exclude the cube-root exponent as a literal clock law at significances from 1626σ to 4762σ – a storage-ring test, an atomic spectroscopy fit, and a four-anchor gravitational-redshift measurement – and none of them is new: the storage-ring bound predates the first edition by a decade, the spectroscopy fit draws on measurements collected over decades, and a dedicated relativistic-clock satellite excluded a comparable exponent before GPS itself ever flew. This second edition confirms that exclusion directly: the real, operationally verified GPS relativistic correction (Ashby 2003, $\approx 38.6\,\mu$s/day) already matches $k = \frac{1}{2}$, reproduced here from first principles to $38.44\,\mu$s/day, and a regression of the originally proposed exponent against real satellite data shows substantial collinearity between the kinematic and gravitational regressors across 145 real GNSS satellites ($r = -0.97$, rising to $r = -0.998$ once the array's one extreme-eccentricity transfer-orbit satellite is set aside), inflating the uncertainty on a satellite-only fit enough that it is a poor way to test either exponent – not, as the data are sometimes read, a test that can never work regardless of sample size. A factor-of-3 arithmetic error in the first edition's own worked example is identified and corrected, along with several further numerical and framing errors in the original surfaced during this revision. What survives: the periodic eccentricity correction, derived in §3, is exactly the one every operational GNSS receiver already applies (ICD-GPS-200, Galileo OS SIS ICD), and QZS-7's extreme transfer-orbit eccentricity remains the sharpest real test case for it. The paper's closing gravitational-wave appendix is reframed on the same terms: its core time-coordinate construction is not physical, and its underlying instinct – that cubic structure matters for radiation – recurs elsewhere in this project only in an untested quantum regime, not as a confirmed physical link; its intuitions about binary-system waveforms are replaced with the actual quadrupole formula, checked against GW150914's published chirp mass and strain to within the expected precision of a leading-order approximation. Every number in this revision is computed through an open, tested backend (`scripts/gnss_model.py`, 14 tests, five checked directly against independently published values) rather than asserted.

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1 Executive Summary

Global Navigation Satellite Systems (GNSS), including GPS, GLONASS, Galileo, and BeiDou, are critical for global navigation, communication, and timing across military, scientific, and civilian domains. The precision and security of these systems rely heavily on accurate timekeeping and the correction of relativistic effects, particularly time dilation.

The first edition of this proposal introduced a cube-root exponent, $(1 - v^2/c^2)^{-1/3}$, in place of the standard Lorentz factor $(1 - v^2/c^2)^{-1/2}$, and reported that 1,000 simulations validated it as more accurate. That claim has not survived independent measurement.

What this edition corrects

Three independent lines of evidence exclude the cube-root exponent as a literal clock law, in both the kinematic and gravitational channels – and, importantly, this is not newly discovered evidence the first edition could not have known about. Most of it predates the first edition, in some cases by decades; the

first edition simply did not check against it:

- A storage-ring measurement of a fast-moving ion (Botermann et al. 2014, Li^+ at $\beta = 0.338$) bounds $|k - \frac{1}{2}| < 1.9 \times 10^{-8}$ – the kinematic channel directly, and ten years old at the time of this revision.
- An independent fit across fourteen one-electron ions, from Mg^{11+} to U^{91+}, drawing on a long-standing compilation of published atomic spectroscopy (Lyman-α and Lamb-shift measurements, not a new atomic-clock experiment), gives $\hat{k} = 0.499962 \pm 0.000102$, excluding $k = \frac{1}{3}$ at 1626σ [CUBIC TIME DIMENSIONS IN QUANTUM MECHANICS, 2ND ED. REVISED, §7.1].
- Four gravitational-redshift anchors give $\hat{k} = 0.50000 \pm 0.00004$, excluding $k = \frac{1}{3}$ at 4762σ [CHRONOWAVE DYNAMICS TEMPORAL ENGINE, MODULE XI].
- Earlier still: NTS-2, the first dedicated relativistic-clock satellite, flew in 1977 – before GPS itself – and already excluded a comparable cube-root-type exponent at 32σ [*Where Cubic Time Lives*, CHRONOWAVE DYNAMICS, §2 BELOW].

This revision keeps the first edition's structure and its genuinely useful machinery – the orbital-mechanics worked calculations, the Taylor-expansion apparatus, the discussion of GNSS architecture – while replacing the central validation claim with what the data actually show. Worked example 1 below gives the clearest single number: the real, decades-old, operationally verified GPS relativistic correction (Ashby 2003) already matches $k = \frac{1}{2}$, not $k = \frac{1}{3}$. The traditional model was never the thing that needed improving; the evidence against replacing it was simply never consulted.

What does survive, and is kept: GNSS clocks genuinely do need a periodic correction from orbital eccentricity, exactly as §3 derives, and that correction is exactly the one every operational GNSS receiver already applies (ICD-GPS-200 §20.3.3.3.3.1; Galileo OS SIS ICD §5.1.4). The first edition's instinct that eccentricity matters for high-precision timing was correct; its instinct that the exponent on the Lorentz factor needed to change was not.

2 Introduction

Relativistic time dilation is a cornerstone of GNSS operations. Time dilation corrections have relied, since the first GPS satellites flew, on the well-established formula $\sqrt{1 - v^2/c^2}$, derived from Einstein's theory of special relativity, combined with the general-relativistic gravitational term. This is not a simplification awaiting replacement; it is the formula that makes GPS work to the precision it already achieves.

Worked example 1: The number this whole proposal is actually about

Ashby (2003), the standard reference on relativity in GPS, gives the net rate offset of a GPS satellite clock relative to a ground clock as approximately +38.6 μs/day: the satellite's weaker gravity speeds its clock up more than its orbital velocity slows it down. Using a representative GPS semi-major axis $a = 26{,}560.6$ km (this proposal's own Module VI satellite database) and Earth's equatorial radius $R_\oplus = 6{,}378.137$ km:

$$\frac{\Delta f}{f} = \frac{GM_\oplus}{c^2}\left(\frac{1}{R_\oplus} - \frac{1}{a}\right) - \frac{v_{\text{sat}}^2}{2c^2}, \qquad v_{\text{sat}} = \sqrt{GM_\oplus/a} = 3.874 \text{ km/s}.$$

Computed directly (not quoted from the literature): $\Delta f/f \times 86400$ s $= 38.44$ μs/day, within the literature's quoted range of the textbook 38.6 μs/day figure. This is the traditional, $k = \frac{1}{2}$ model, checked against one of the most repeatedly verified numbers in applied relativity. Try it: the velocity-only (special-relativistic) piece alone gives -7.21 μs/day, matching the commonly quoted -7 μs/day GPS

figure on its own [BACKEND: SCRIPTS/GNSS_MODEL.PY, TEST_GNSS_MODEL.PY; OF ITS 14 TESTS, 5 CHECK DIRECTLY AGAINST INDEPENDENTLY PUBLISHED VALUES INCLUDING THIS ONE].

The first edition proposed replacing this exponent with $(1 - v^2/c^2)^{-1/3}$, on the premise that a different exponent would reduce positional errors. Worked example 2 shows what that exponent actually predicts for the same satellite, so the two claims can be compared directly rather than taken on faith.

Worked example 2: What the cubic exponent predicts, computed the same way

For the same GPS satellite, $v = 3.874$ km/s, $\beta^2 = 1.6698 \times 10^{-10}$:

$$\left(1 - \beta^2\right)^{1/2} - 1 \to -7.213\ \mu\text{s/day}, \qquad \left(1 - \beta^2\right)^{1/3} - 1 \to -4.809\ \mu\text{s/day},$$

a split of 2.40 μs/day between the two models – small compared to a day, but enormous compared to the $\sim$20–30 ns of timing precision GNSS actually requires. In practice, a *constant* per-satellite clock-rate error like this one would not, by itself, show up as a dramatic position error: GPS continuously recalibrates each satellite's clock against ground control through the broadcast navigation message's own drift parameters, and a steady model bias would simply be absorbed into those correction terms. The real discriminating test is a direct measurement of the clock rate itself, not a downstream position-error argument, and that test already exists and predates this proposal by decades: NTS-2 (1977), a dedicated relativistic-clock satellite flown before GPS itself, measured the predicted offset directly to about 1% and already excludes a comparable cube-root-type exponent at 32σ [*Where Cubic Time Lives*, CHRONOWAVE DYNAMICS]. GPS has run on the traditional, $k = \frac{1}{2}$ model since its first operational satellite launched in 1978, reaching full operational capability in 1995, and nothing in decades of GNSS clock-correction telemetry has ever required the 2.40 μs/day shift this proposal predicts [THIS IS THE REAL-WORLD NULL RESULT THE 1,000-SIMULATION CLAIM SHOULD HAVE BEEN CHECKED AGAINST].

3 Theoretical Foundation

3.1 Traditional Time Dilation

Special relativity predicts time dilation as an object's velocity approaches the speed of light, exactly, not approximately:

$$\Delta t' = \Delta t \sqrt{1 - \frac{v^2}{c^2}}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

where Δt is the coordinate time interval measured by a stationary (ground) observer, $\Delta t'$ the proper time elapsed on the moving clock, v the object's velocity, and c the speed of light. This is not a truncated approximation awaiting a better one – the square root is the exact closed form, valid at any speed below c. This formula is the standard for correcting GNSS timing signals, and worked example 1 above shows it is correct to the precision GNSS actually achieves.

3.2 Cubic Time Dilation

The first edition proposed replacing the exponent $-\frac{1}{2}$ with $-\frac{1}{3}$:

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-1/3}.$$

What this edition corrects

The first edition argued that this exponent change "retains more of the [Taylor] series expansion's higher-order terms" than the traditional model, and is therefore more precise. Worked example 3 shows this reasoning does not hold: the cubic model is not the traditional model's series carried further. It is a different function, with a different, and wrong, leading-order coefficient.

3.3 Mathematical Derivation and Justification

The exact Lorentz factor expands, for $v \ll c$, as

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = 1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \frac{5}{16}\frac{v^6}{c^6} + \cdots,$$

and the cubic proposal's own function expands as

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-1/3} = 1 + \frac{1}{3}\frac{v^2}{c^2} + \frac{2}{9}\frac{v^4}{c^4} + \cdots.$$

Worked example 3: "More terms of the series" is not what changed – and an arithmetic error in the original, caught by recomputing rather than copying

The first edition's own worked comparison, at $v = 3 \times 10^3$ m/s, reported $\gamma \approx 1 + 1.5 \times 10^{-10}$ for the traditional model (first series term only) and $\gamma_{\text{cubic}} \approx 1 + 1 \times 10^{-10}$ for the cubic model (two series terms), then presented the smaller cubic number as evidence of superior precision. Recomputed directly, with $\beta^2 = (3 \times 10^3/c)^2 = 1.00139 \times 10^{-10}$:

$$\gamma - 1 = \tfrac{1}{2}\beta^2 = 5.007 \times 10^{-11}, \qquad \gamma_{\text{cubic}} - 1 = \tfrac{1}{3}\beta^2 + \tfrac{2}{9}\beta^4 = 3.338 \times 10^{-11}.$$

Both of the first edition's own quoted figures are too large by a factor of $2.996 \approx 3$ – an arithmetic slip in the original worked example, not a modeling choice, and one this revision's backend caught immediately by recomputing rather than retyping the original numbers [VERIFIED INDEPENDENTLY: SCRIPTS/TEST_GNSS_MODEL.PY]. The corrected comparison makes the real point more clearly, not less: these are not the same series carried to different depths, but two different functions, and the traditional model's exact value at this speed, $\sqrt{1-\beta^2}^{-1} - 1 = 5.0069253 \times 10^{-11}$, matches its own one-term truncation to better than eight significant figures, because $\beta^4 \sim 10^{-21}$ is negligible here, not because the truncation was inadequate. The cubic model's $\frac{1}{3}$ leading coefficient is not "more precise than $\frac{1}{2}$" – it is a 33% change to the *leading* term of a different physical law, which no number of additional series terms can repair, because the mismatch lives at the very first correction, $O(v^2/c^2)$, where both models already disagree with each other at the few-hundredths-of-a-part-per-billion level that GNSS measures continuously (worked example 2).

The traditional model is not an approximation that cubic "improves upon by keeping more terms." It is the exact relativistic prediction, confirmed to the precision shown in worked example 1. The cubic model's error is structural: a wrong leading-order coefficient, not a missing higher-order correction.

3.4 The Periodic Eccentricity Correction

Worked example 1's 38.44 μs/day is the *constant* part of a GNSS clock's relativistic offset. For an eccentric orbit there is also a part that oscillates once per orbit, because both the satellite's

speed and its distance from Earth vary with orbital phase – fastest and lowest at perigee, slowest and highest at apogee. To first order, the proper-time rate of a clock obeying a general dilation exponent k (so $k = \frac{1}{2}$ is the standard, GR-correct case and $k = \frac{1}{3}$ is this proposal's cubic case) relative to a distant coordinate clock is

$$\frac{d\tau}{dt} = 1 - (1+2k)\,\frac{\mu}{rc^2} + k\,\frac{\mu}{ac^2},$$

where $\mu = GM_\oplus$, r is the satellite's instantaneous geocentric distance, and a its semi-major axis. Writing $r = a(1 - e\cos E)$ in terms of the eccentric anomaly E, the $1/r$ term splits into a constant piece (already folded into the net rate above) and a piece proportional to $e\sin E$ that averages to zero over one orbit but is not zero at any given instant. Integrating that piece gives the periodic correction

$$\Delta t_r(k) = -(1+2k)\sqrt{\mu a}\,e\sin E/c^2.$$

At $k = \frac{1}{2}$, $(1+2k) = 2$, recovering the correction every operational GNSS receiver already applies every update cycle (ICD-GPS-200 §20.3.3.3.3.1; Galileo OS SIS ICD §5.1.4):

$$\Delta t_r\left(\tfrac{1}{2}\right) = -2\sqrt{\mu a}\,e\sin E/c^2.$$

At $k = \frac{1}{3}$, the cubic proposal's own exponent, $(1+2k) = \frac{5}{3}$ instead of 2 – a coefficient ratio of $\frac{5}{6}$ against the operational value, the split computed per satellite in Appendix A and reproducing Module VI's own published 263.0 ns (QZS-7) and 36.3 ns (QZS-2) figures for this quantity.

4 Validation and Results

What this edition corrects

The first edition reports "1,000 simulations" validating the cubic model across nine scenario categories (satellite systems, atmospheric conditions, orbital scenarios, anomalies, gravitational effects, cross-platform integration, velocity extremes, maintenance, and ground-observer variation), but presents no methodology, no data, and no quantitative result for any of them – only the claim that each category showed "consistent accuracy improvements." There is nothing in the original text that can be independently checked. This edition replaces the claim with an actual computation, and the computation explains why a test like the one described would show "consistent improvement" for essentially any exponent, including the wrong one.

Worked example 4: Why a satellite-only test is a poor way to fit either exponent, computed independently

A regression that tries to fit the dilation exponent k from satellite data alone regresses the clock-rate offset against two candidate predictors: $\beta^2 = v^2/c^2$ (the kinematic term) and Φ/c^2 (the gravitational term). If these two predictors are strongly correlated across the satellites being tested, a regression can still fit both coefficients, but with inflated uncertainty – making it an unreliable way to pin down k without either more statistical power or a predictor that is not collinear with the others, not an argument that no amount of data could ever work in principle (worked example 2 already shows real operational data distinguishing the exponents directly, through the periodic eccentricity term, without needing this regression at all).
Computed here independently, using vis-viva speed at perigee (which depends on each satellite's own eccentricity, not just its semi-major axis) against gravitational potential at the same point, across all 145 satellites in the Chronowave Dynamics Module VI database:

$$r(\beta^2, \Phi) = -0.974, \qquad \text{VIF} = 19.2.$$

This is consistent, in sign and order of magnitude, with Module VI's own live-computed collinearity panel for the same satellite set [MODULE VI, "THE DEGENERACY, MEASURED RATHER THAN ASSERTED"] – though Module VI's figure uses real-time orbital phase and therefore drifts by the hour, while the value above uses only each satellite's fixed semi-major axis and eccentricity and is reproducible on demand. The -0.974 figure is itself a *conservative* reading: it is driven almost entirely by QZS-7, the one satellite in the array on a highly eccentric transfer orbit. Excluding that single satellite, the remaining 144 – essentially all of them on near-circular operational orbits – are very nearly degenerate, $r = -0.998$ (VIF ≈ 289, inflating the standard error on a fitted exponent by a factor of about 17, versus about 4.4 with QZS-7 included). In other words, operational GNSS satellites alone are close to a single collinear population; it is specifically the one atypical, non-operational orbit in the array that keeps the regression even as well-conditioned as VIF 19. Module VI's own further finding, that adding ground-based clocks (at a fixed a no satellite occupies) sharply reduces this collinearity, is the practical fix: Module VI demonstrates this live, as a regression row whose coefficient stays stable once ground stations are added and drifts further from the true value as synthetic clock noise is raised when they are left out, even though it continues to pick the correct model through the noise levels reachable in that demonstration. The eccentricity-based test derived in §3.4, not a satellite-only regression, is the discriminator this paper actually relies on; the regression above explains why a cruder test could look inconclusive, not why the question itself is unanswerable.

The one discriminating observable that does separate the models without needing ground clocks is the periodic eccentricity correction derived in §3.4 and worked out per satellite in Appendix A: because it depends on $e \sin E$ rather than on β^2 or Φ alone, it is not degenerate with either, and it is the same correction every operational GNSS receiver already applies and checks continuously (ICD-GPS-200 §20.3.3.3.3.1).

5 Mars Missions, Near-Space, and Deep-Space Applications

The first edition argues that the cubic model's "higher-order corrections" make it particularly suited to Mars missions, near-space operations, and deep-space exploration, where velocities and gravitational environments vary more than they do for Earth-orbiting GNSS satellites.

What this edition corrects

The argument does not depend on which exponent is correct – it only argues that *some* relativistic correction matters more at higher speeds and larger gravitational-potential changes, which is true of the correct ($k = \frac{1}{2}$) model exactly as it would be of any other. Nothing in this section is specific to the cube-root exponent, and nothing here is corrected by abandoning it; the traditional model already extends to these regimes, since $\sqrt{1 - v^2/c^2}$ is an exact closed form at any sub-light speed, not an approximation valid only near Earth orbital velocities.

What is worth keeping is the observation itself: deep-space missions do accumulate larger relativistic corrections than Earth-orbit GNSS, because they spend more time at higher $|\Phi|$ and, for some trajectories, higher v. The Mars Reconnaissance Orbiter and other deep-space missions already apply relativistic corrections derived from the standard (not cubic) model in their navigation solutions [THIS IS STANDARD JPL/NASA DEEP-SPACE NAVIGATION PRACTICE, NOT A NOVEL APPLICATION OF THIS PAPER'S PROPOSAL]; the genuine opportunity for future work is extending the same eccentricity-correction formalism from §3 to highly elliptical interplanetary transfer orbits, where e is far larger than any GNSS satellite's and the periodic term correspondingly larger – not in replacing the exponent that governs the constant term.

6 Implementation Strategy

What this edition corrects

The first edition's implementation strategy is a seven-step rollout plan – pilot testing, compatibility assessment, further theoretical research, computational-overhead evaluation, stakeholder engagement, input-data sensitivity, and extreme-environment monitoring – written entirely in the future tense, with no numbers, for a model change it proposes to roll out onto live GNSS infrastructure. That plan is moot: §3–4 already show the proposed change is excluded, and the direct pilot test this plan called for already happened, deliberately, before GPS even flew: NTS-2 (1977) measured the predicted relativistic offset directly and already excludes a comparable cube-root-type exponent at 32σ (worked example 2). GPS itself has run on the traditional model since its first satellite launched in 1978. What follows keeps the plan's structure but is honest about which parts are real engineering concerns worth keeping for *any* future proposed change to GNSS timing models, independent of this one's outcome.

6.1 Empirical Validation

Moot for this proposal specifically – the empirical validation already exists, continuously, in the form of every GNSS fix made since 1978, and it confirms the traditional model (worked example 1). The general principle survives: any future proposed change to a GNSS timing model should be checked against the discriminating observable identified in §4 (the eccentricity periodic term), which is not subject to the collinearity that makes a satellite-only regression on the constant-rate term an unreliable way to fit a dilation exponent (worked example 4).

6.2 Compatibility with Existing Systems

Not applicable: there is nothing to integrate. The one subsection here worth keeping as general guidance is the instinct itself – any real change to operational GNSS timing software needs a compatibility and rollback plan – which remains true for whatever future correction, if any, is eventually proposed and actually survives validation.

6.3 Higher-Order Effects

This is the one subsection with a real, surviving question: whether the periodic eccentricity correction (§3) interacts with frame-dragging or other second-order GR effects at a level GNSS precision can detect. It does not, at current precision – frame-dragging's effect on a GPS-altitude clock is many orders of magnitude below the eccentricity term computed in Appendix A – but this is a genuine, checkable question, unlike the rest of this section, and is worth keeping open for future higher-precision systems.

6.4 Computational Overhead

Not applicable to a model that is not being implemented. The periodic eccentricity correction already in operational use (ICD-GPS-200) costs one sine evaluation per satellite per update cycle, already budgeted into every existing GNSS receiver; nothing in this paper changes that cost.

6.5 Adoption and Stakeholder Engagement

Removed. A communication strategy for adopting a model that independent measurement has excluded would not be engineering, and this edition does not recommend pursuing one.

6.6 Sensitivity to Input Data

The one place this concern is genuinely sharp: Module VI's own documentation of QZS-7, whose current transfer-orbit eccentricity (e=0.721) makes its eccentricity-correction amplitude about seven times larger than its own constellation sibling QZS-2's (e=0.076), and temporary – it is expected to shrink once QZS-7 reaches its intended operational quasi-zenith orbit, which, like QZS-2's, will still have an eccentricity of roughly 0.08, not the near-circular orbit a GPS satellite flies. Any discriminator built on eccentricity alone needs to flag transfer orbits as a different population, not an outlier.

6.7 Impact on Extreme Environments

Covered in §5: the genuine content here is that interplanetary transfer orbits have much larger eccentricities than any GNSS satellite, making the periodic correction term correspondingly larger and more important to get right – not that a different constant-term exponent is needed.

7 Conclusion

The first edition proposed a cube-root exponent for relativistic time dilation in GNSS, arguing it offered superior accuracy over the traditional square-root (special-relativistic) factor. That central claim does not survive: three independent lines of evidence – a storage-ring test, an atomic spectroscopy fit, and a four-anchor gravitational-redshift measurement, together with an even earlier direct clock-rate test – exclude the cube-root exponent as a literal clock law at significances from 1626σ to 4762σ (§1), none of it newly available evidence, and the real-world system this paper proposed to improve already runs, and has run for decades, on the exponent this proposal wanted to replace (worked example 1). The "1,000 simulations" validation claim is not independently checkable as written, and the quantitative mechanism this revision could reconstruct – substantial collinearity between the kinematic and gravitational regressors across a satellite-only sample, worse still once the one atypical transfer-orbit satellite in the array is excluded – explains why a satellite-only constant-rate regression is a poor tool for fitting either exponent, not why the question is unanswerable (§4).

What survives is narrower, and real. GNSS clocks do carry a periodic correction from orbital eccentricity, derived honestly in §3.4 and already the one every operational system applies; that correction, not a change to the constant-term exponent, is where eccentricity-driven precision actually lives, and QZS-7's extreme, temporary eccentricity (§6) is the sharpest real test case for it. The orbital-mechanics worked calculations in Appendices A–B, redone here against the verified backend and Module VI's own published satellite database, are kept because the underlying approach – Kepler's laws, the Taylor-series apparatus, per-satellite worked examples – was sound; several of the first edition's own numerical results along the way were not (a mislabeled semi-major axis for non-GPS satellites, a sign error in its own narrative about which direction the net effect runs,

and at least one order-of-magnitude slip in a quoted value, documented in §3 and Appendix A), and this revision corrects those rather than carrying them forward unexamined.

This is not a story of a proposal that failed outright. It is a story of a proposal whose genuine instinct – that high-precision timing systems deserve careful relativistic treatment – pointed at a real and already-solved problem, and whose specific mechanism for solving it differently turned out, on the evidence, not to be needed.

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A Per-satellite worked calculations: first and second order

What this edition corrects

The first edition names satellites by generic block/type labels (e.g. "GLONASS-K1," "GALILEO-FOC FM15," "GPS BIIA-10") that do not correspond one-to-one to individual catalogued satellites, and derives each one's orbital elements by hand, separately, at first, second, and fourth order – roughly thirty individual hand calculations in this appendix alone, independent of the main text's worked example 3 (a separate toy calculation at $v = 3 \times 10^3$ m/s), which was already shown to carry its own factor-of-3 arithmetic error. Checking the per-satellite calculations directly against the original PDF turns up further, separate errors: its own hand-worked example for a Galileo-type satellite uses a semi-major axis of 26,560 km – GPS's value, not Galileo's real ≈ 29,600 km or BeiDou's real ≈ 27,906 km, both of which it applies to the wrong satellites elsewhere in the same table – and its own stated gravitational-term result for that calculation (2.793×10^{-10}) is about 1.8× too large even given the inputs it used; separately, one quoted result elsewhere in the appendix is off by a full order of magnitude ($1 - 8.35 \times 10^{-12}$ where the calculation's own inputs give $1 - 8.35 \times 10^{-11}$), and the appendix's prose describes the net relativistic effect as leaving satellite clocks "slightly less than one" relative to the ground – i.e. running slow – when the real, operationally measured net effect (worked example 1) is that they run fast. This revision instead uses one real, individually catalogued satellite per constellation, with orbital elements taken directly from Module VI's own published 145-satellite database, and computes every entry in Table 1 through the verified backend (`scripts/gnss_model.py`) rather than by hand.

Worked example 5: NAVSTAR 43, worked by hand and then checked against the backend

NAVSTAR 43 (USA 132), a GPS Block IIR satellite, has semi-major axis $a = 26{,}560.6$ km and eccentricity $e = 0.010485$ (Module VI database). Its circular-equivalent orbital speed is $v = \sqrt{GM_\oplus/a} = 3.8739$ km/s, giving $\beta^2 = 1.6698 \times 10^{-10}$.

$$\begin{aligned}
\text{Traditional, 1st order:} \quad & \gamma - 1 = \tfrac{1}{2}\beta^2 = 8.3489 \times 10^{-11} \\
\text{Traditional, exact:} \quad & \gamma - 1 = (1-\beta^2)^{-1/2} - 1 = 8.3489 \times 10^{-11} \\
\text{Cubic, 2nd order:} \quad & \gamma_{\text{cubic}} - 1 = \tfrac{1}{3}\beta^2 + \tfrac{2}{9}\beta^4 = 5.5659 \times 10^{-11} \\
\text{Cubic, exact:} \quad & \gamma_{\text{cubic}} - 1 = (1-\beta^2)^{-1/3} - 1 = 5.5659 \times 10^{-11}
\end{aligned}$$

Unlike the first edition's worked example, the first-order and exact values agree here to the precision shown, because this satellite's $\beta^4 \sim 2.8 \times 10^{-20}$ is, correctly, negligible. The split between the two models is 2.40 μs/day (worked example 2); the eccentricity-correction split at this satellite's $e = 0.010485$, reproducing Module VI's own formula, is 4.00 ns at maximum, against what that same formula and database give for QZS-2 ($e = 0.075525$, $a = 42{,}164.6$ km, a near-geosynchronous orbit, 36.3 ns) and QZS-7 ($e = 0.720983$, $a = 24{,}260.6$ km, a transfer orbit, 263.0 ns) – Module VI computes these live per satellite rather than printing them, so they are reproduced here, not quoted. The formula scales as $e\sqrt{a}$, not e alone, and checking both ratios against this satellite confirms it exactly: QZS-2's e is 7.20× NAVSTAR 43's, and its larger a contributes a further $\sqrt{42{,}164.6/26{,}560.6} = 1.26\times$, predicting a 9.08× larger split – the computed ratio is 9.08, not approximately so but to three significant figures. QZS-7's smaller a works the other way ($\sqrt{24{,}260.6/26{,}560.6} = 0.9557\times$) against its 68.76× larger e, predicting 65.7×, matching the computed ratio exactly.

Two things are worth naming explicitly. First, the constant-term split (last column) is nearly identical across these five representative satellites, 2.2–2.6 μs/day, despite a 69-fold range in eccentricity across the five constellations shown – because the constant-term split depends on orbital speed and altitude, not eccentricity, and these particular satellites (QZS-7's current transfer orbit aside) all occupy similar MEO altitude bands. The full 145-satellite database also includes a GEO/IGSO/QZSS/NavIC population at roughly 42,165 km, whose constant-term split is cor-

Table 1: Verified per-satellite comparison, one representative per constellation, computed via `scripts/gnss_model.py` against Module VI's own published orbital elements.

Sat.	a (km)	e	v (km/s)	Trad. $\gamma-1$	Cubic $\gamma-1$	Split
NAVSTAR 43	26,560.6	0.010485	3.8739	8.349e−11	5.566e−11	2.4045
COSMOS 2434	25,508.1	0.000118	3.9530	8.693e−11	5.796e−11	2.5037
GSAT0101	29,600.1	0.000135	3.6696	7.492e−11	4.994e−11	2.1576
BEIDOU-3 M1	27,906.1	0.000824	3.7794	7.946e−11	5.298e−11	2.2885
QZS-7	24,260.6	0.720983	4.0534	9.140e−11	6.094e−11	2.6324

NAVSTAR 43 = GPS; COSMOS 2434 = GLONASS-M; GSAT0101 = Galileo-PFM; QZS-7 = transfer orbit. Split in μs/day.

respondingly smaller, around 1.5 μs/day; the near-constancy shown here is a property of the MEO shell specifically, not of GNSS satellites in general, and is a separate fact from the predictor collinearity discussed in §4, which is driven by near-circularity (low eccentricity), not by a narrow altitude range. Second, QZS-7's eccentricity does not move the constant-term split meaningfully (2.63 vs. 2.40 μs/day for GPS) but moves the *periodic* eccentricity split enormously (worked example 5 and §6) – confirming that eccentricity is the genuinely discriminating lever, not orbital speed.

B Per-satellite worked calculations: fourth order

Extending Table 1 to fourth order in β^2 – through the β^8 term – changes nothing observable. At GNSS orbital speeds, $\beta^2 \sim 10^{-10}$, so the fourth-order term itself is of order $\beta^8 \sim 10^{-39}$–10^{-40}, computed via `taylor_expand(beta2, k, order=4)`; converted to a time scale ($\times$86,400 s/day), that term alone is of order 10^{-35} s/day, more than thirty orders of magnitude below GNSS's $\sim$ 20–30 ns timing precision. The residual left after truncating at fourth order is smaller still, of order $\beta^{10} \sim 10^{-50}$ – already far below the $\sim 10^{-16}$ noise floor of standard double-precision arithmetic, which is why a direct floating-point subtraction of the fourth-order and exact values returns exactly zero (or floating-point noise at the 10^{-16} level) rather than a meaningful number at this scale; the true smallness of the term is an analytic fact about the size of β^8, not something floating-point subtraction can resolve on its own. The first edition's fourth-order expansions (originally worked by hand for BEIDOU-3 M14, GLONASS-K1, GALILEO-FOC FM15, and BEIDOU-3 M18) are reproduced here as a null result: going beyond second order adds no information that GNSS can measure, for either model, at any real satellite's speed. The appendix is kept, in compressed form, because the first edition's own effort to carry the expansion further was a reasonable instinct to check – it is a useful, checkable fact that the answer is no, not merely an assumption.

C The gravitational-wave coherence appendix: what it was reaching for

What this edition corrects

The first edition's final 79 pages (from p. 34 of 112) leave the GNSS topic entirely and sketch a different idea: that time itself might have a "signature" – linear, cubic, or otherwise – and that gravitational waves might be "coherent" or "decoherent" depending on how these signatures align between observers. The core construction, `cubicTime[t_] := t^3` composed additively with linear time into a "composite time signature," is not a physical model: it cubes the time *coordinate* itself, which has no

coordinate-independent meaning and no connection to any measurable quantity (a coordinate can be relabeled $t \to t^3$ freely; nothing observable changes by doing so, which is a basic feature of general covariance, not a loophole). But the underlying instinct – that something cubic might matter for radiation – is not unreasonable to have chased. Worked example 6 identifies, independently of this appendix, the one place in this project's own work where a structurally similar exponent has actually been derived for a real observable – though as a theoretical, not yet experimentally tested, prediction, not a confirmed result.

Worked example 6: Where a cube-root exponent actually shows up in radiation, verified elsewhere in this project

The companion note *Where Cubic Time Lives* (Chronowave Dynamics, `Research/Temporal/cubic_time/`) finds, independently of this paper, that the exponent $(1-\beta^2)^{-1/3}$ *does* recur in real physics – not as a clock-rate law (excluded, §1), but in the quantum limit of radiated power from an accelerating charge in a fixed strong field, whose exponent is set by the Airy structure of that emission. The note is explicit that this recurrence is a structural coincidence, not a consequence of any argument either project makes, and that the regime where it would show up has not been reached experimentally; it is not a confirmed physical link, and the note separately develops radiation reaction (the jerk, d^3x/dt^3) as a related but distinct topic, also unconnected to a literal clock-rate exponent. This is still a precise and checkable claim, entirely different from "time has a cubic signature": it says that a specific, as-yet-untested observable – radiated power, not clock rate, not the time coordinate – carries this exponent in one identified theoretical regime. The first edition's gravitational-wave appendix was reaching for a connection in this neighborhood (cubic structure, radiation, extreme fields) but attached it to an undefined coordinate relabeling instead of the one place in this project's own work where a structurally similar exponent has actually been derived – itself still an unconfirmed prediction, not a verified result.

The correct redirection for this appendix, then, is not to abandon the intuition that cubic structure and gravitational radiation might be connected, but to point it at the actual physics of gravitational-wave generation: the quadrupole formula, worked out in §D, and the jerk and radiation-reaction framework developed, as a separate theoretical topic, in *Where Cubic Time Lives*.

D Gravitational waves and temporal rhythms

What this edition corrects

"Coherence" and "decoherence" are given precise, well-established meanings in real gravitational-wave astronomy, and neither matches how the first edition uses the terms. In LIGO and Virgo data analysis, a signal is detected by *matched filtering*: correlating the detector strain against a bank of theoretical waveform templates, each one a phase-coherent prediction (from post-Newtonian theory, numerical relativity, or both) of exactly how a binary's strain evolves as it inspirals and merges. "Coherence" here means the template's phase tracks the true signal's phase over many cycles; loss of that tracking – from detector noise, an imprecise template, or genuine deviation from general relativity – degrades the matched-filter signal-to-noise ratio. It is a property of the *analysis*, relative to a specific template and a specific noisy detector, not an intrinsic property of "temporal rhythms" aligning between observers or regions of spacetime.

Worked example 7: The actual physics, checked against a real detection

The leading-order (Newtonian quadrupole) strain amplitude from a circular-orbit compact binary is

$$h = \frac{4}{d}\left(\frac{G\mathcal{M}_c}{c^2}\right)^{5/3}\left(\frac{\pi f_{\rm gw}}{c}\right)^{2/3}, \qquad \mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1+m_2)^{1/5}},$$

where $\mathcal{M}_c$ is the chirp mass and f_{gw} the gravitational-wave frequency (twice the orbital frequency). For GW150914 (Abbott et al. 2016), two black holes of 36 and 29 $M_\odot$ merging at 410 Mpc, evaluated near $f_{\mathrm{gw}} = 150$ Hz: $\mathcal{M}_c = 28.1\,M_\odot$ (the published chirp mass, $\approx 28\,M_\odot$, follows directly from the published component masses rather than being independently fit here) and $h = 2.13 \times 10^{-21}$ – about 2.1× LIGO's quoted peak strain of $\approx 1.0 \times 10^{-21}$, consistent with the known limitation that the leading Newtonian quadrupole formula is not expected to be precise this close to merger, where orientation, detector response, and the transition to full numerical relativity all matter [BACKEND: `BINARY_INSPIRAL_STRAIN_AMPLITUDE()`, `TEST_QUADRUPOLE_STRAIN_MATCHES_GW150914_ORDER_OF_MAGNITUDE`]. This is the real content the first edition's "Rhythmic Patterns of Time" and "Gravitational Wave Coherence" subsections were gesturing toward: a genuine, checkable formula connecting source parameters to detector strain, no ad hoc time-signature language required.

What does carry over honestly from the first edition's instinct: binary systems in stable, near-circular orbits do produce clean, well-characterized ("coherent," in the ordinary sense of periodic and predictable) waveforms, exactly as a two-body Keplerian system's quadrupole moment oscillates predictably; eccentric or precessing systems produce messier waveforms that are genuinely harder to match-filter. That much of the music-theory-adjacent intuition (clean rhythm vs. chaotic rhythm) survives as a loose analogy to *detectability*, not as a claim about time itself having structure.

E Observational implications and the simulation script

The first edition's closing material proposes comparing simulated waveforms against real LIGO/Virgo data via cross-correlation, and sketches a `FullPyScriptGrav` script toward that end. Cross-correlation against real strain data is, in fact, close to the right idea – matched filtering against real detector output *is* how LIGO/Virgo actually find signals – but the sketch itself generates its comparison signal from the undefined "coherent/decoherent time signature" construction of §§C–D, not from a real waveform model, so a genuine cross-correlation against real strain data would correlate against noise.

What this edition corrects

Replacing the undefined signal generator with the quadrupole formula of worked example 7 turns this from a speculative sketch into a real, if leading-order, gravitational-wave template – publicly available LIGO strain data (GWOSC, the Gravitational Wave Open Science Center) could then genuinely be cross-correlated against it, and the result would be a real, if approximate, signal-processing exercise rather than a correlation against an undefined quantity. Building that pipeline fully is beyond the scope of this revision, which is about the GNSS proposal; it is recorded here as the concrete, well-posed version of what the first edition's final appendix was trying to do.